

Lecture Scribe: CSE 400 Lecture 20

Topic: Linear Transformations and Expectations of Multiple RVs

Scribe: Fagun Rathod - AU2440111

March 25, 2026

1 Random Vector Representation

1.1 Definitions and Notation

- **Random Vector:** For random variables $X_1, X_2, \dots, X_N$, they are represented as a vector $\underline{X}$.
- **Vector Form:**

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- **Purpose:** Provides a compact representation of multiple random variables.

2 Statistical Representation: Mean Vector

2.1 Definitions and Notation

- **Mean Vector Definition:** The mean vector of $\underline{X}$ is $\underline{\mu}_X = E[\underline{X}]$.
- **Notation:**

$$\underline{\mu}_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

2.2 Assumptions / Conditions

- Each element in the mean vector corresponds to the mean of one individual random variable.
- It represents the average value of the random vector.

3 Statistical Representation: Correlation Matrix

3.1 Definitions and Notation

- **Correlation Matrix (R_{XX}):** Defined as $R_{XX} = E[\underline{X}\underline{X}^T]$.
- **Element Definition:** $R_{XX}(i, j) = E[X_i X_j]$ for $i, j \leq N$.
- **Matrix Structure:**

$$R_{XX} = \begin{bmatrix} E[X_1 X_1] & E[X_1 X_2] & \dots & E[X_1 X_N] \\ E[X_2 X_1] & \dots & \dots & E[X_2 X_N] \\ \vdots & \vdots & \vdots & \vdots \\ E[X_N X_1] & E[X_N X_2] & \dots & E[X_N X_N] \end{bmatrix}_{N \times N}$$

3.2 Assumptions / Conditions

- R_{XX} is a symmetric matrix.

4 Statistical Representation: Covariance Matrix

4.1 Definitions and Notation

- **Covariance Matrix (C_{XX}):** For a random vector $\underline{X}$, $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.
- **Practical Structure:**

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \dots \\ Cov(X_2, X_1) & Var(X_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

4.2 Assumptions / Conditions

- **Symmetry:** $Cov(X_i, X_j) = Cov(X_j, X_i)$, making the matrix a "mirror" of itself.

5 Linear Transformations of Random Vectors

5.1 Definitions and Notation

Linear Model: $\underline{Y} = \underline{A}\underline{X} + \underline{b}$

- $\underline{A}$: Transformation matrix (scales and rotates data).
- $\underline{b}$: Constant vector (shifts/translates the center).
- $\underline{X}$: Input random vector.

Property	Transformed Result
Mean	$\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$
Covariance	$\underline{C}_{YY} = \underline{A}\underline{C}_{XX}\underline{A}^T$
Correlation	$\underline{R}_{YY} = \underline{A}\underline{R}_{XX}\underline{A}^T + \underline{A}\underline{\mu}_X\underline{b}^T + \underline{b}\underline{\mu}_X^T\underline{A}^T + \underline{b}\underline{b}^T$

5.2 Main Results / Theorems

5.3 Proofs / Derivations

Mean Transformation Derivation

1. $\underline{\mu}_Y = E[\underline{Y}]$
2. Substitute $\underline{Y} = \underline{A}\underline{X} + \underline{b}$: $\underline{\mu}_Y = E[\underline{A}\underline{X} + \underline{b}]$
3. Apply linearity: $E[\underline{A}\underline{X}] + E[\underline{b}]$
4. Result: $\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$

Correlation Matrix Transformation Derivation

1. $\underline{R}_{YY} = E[\underline{Y}\underline{Y}^T] = E[(\underline{A}\underline{X} + \underline{b})(\underline{A}\underline{X} + \underline{b})^T]$
2. Expansion: $\underline{R}_{YY} = E[\underline{A}\underline{X}\underline{X}^T\underline{A}^T + \underline{A}\underline{X}\underline{b}^T + \underline{b}\underline{X}^T\underline{A}^T + \underline{b}\underline{b}^T]$
3. Applying Expectation: $\underline{R}_{YY} = \underline{A}E[\underline{X}\underline{X}^T]\underline{A}^T + \underline{A}E[\underline{X}]\underline{b}^T + \underline{b}E[\underline{X}^T]\underline{A}^T + \underline{b}\underline{b}^T$
4. Result: $\underline{R}_{YY} = \underline{A}\underline{R}_{XX}\underline{A}^T + \underline{A}\underline{\mu}_X\underline{b}^T + \underline{b}\underline{\mu}_X^T\underline{A}^T + \underline{b}\underline{b}^T$

5.4 Worked Examples

Example 1: Numerical Linear Transformation

Given: $\underline{Y} = \underline{A}\underline{X}$ where:

$$\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad \underline{R}_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

- (a) Mean vector of $\underline{Y}$:

$$\underline{\mu}_Y = \underline{A}\underline{\mu}_X = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

- (b) **Correlation matrix of $\mathbf{Y}$:** $\underline{R}_{YY} = \underline{A}\underline{R}_{XX}\underline{A}^T$

$$\text{Step 1: } \underline{A}\underline{R}_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

$$\text{Step 2: } \underline{R}_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

- (c) **Covariance matrix of $\mathbf{Y}$:** $\underline{C}_{YY} = \underline{R}_{YY} - \underline{\mu}_Y \underline{\mu}_Y^T$

$$\underline{\mu}_Y \underline{\mu}_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

$$\underline{C}_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

6 Multivariate Gaussian Distribution

6.1 Definitions and Notation

- **Univariate (1D):**

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- **Joint Gaussian (2D):**

$$f_{XY}(x, y) = \frac{\exp\left\{-\frac{1}{2(1-\rho_{XY}^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY} \frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}}$$

- **N-Dimensional Vector PDF:**

$$f_{\underline{X}}(\underline{\mathbf{x}}) = \frac{1}{\sqrt{(2\pi)^N \det(\underline{C}_{XX})}} \exp\left(-\frac{1}{2}(\underline{\mathbf{x}} - \underline{\mu}_X)^T \underline{C}_{XX}^{-1}(\underline{\mathbf{x}} - \underline{\mu}_X)\right)$$

6.2 Main Results / Theorems

- **Uncorrelated Gaussian RVs:** If jointly Gaussian RVs are mutually uncorrelated ($\text{Cov}(X_i, X_j) = 0$), the covariance matrix $\underline{C}_{XX}$ is diagonal.
- **Factorization:** The joint PDF factorizes:

$$f_{\underline{X}}(\underline{\mathbf{x}}) = \prod_{n=1}^N \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$